

GEELONG, Australia

WMO: 948570

Lat: 38.1736S

Lon: 144.3764E

Elev: 44

StdP: 14.67

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.7	38.8	32.7	27.2	49.9	34.8	29.7	48.0	27.8	54.2	25.5	52.9	3.8	290	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.8	93.2	66.0	87.5	65.4	81.9	64.3	70.2	81.7	68.4	79.1	66.9	77.0	14.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.2	100.1	72.2	65.4	93.9	71.2	63.7	88.5	69.9	34.2	82.0	32.7	79.2	31.4	77.1	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	22.6	20.1	31.6	104.3	2.2	4.5	30.0	107.6	28.8	110.2	27.5	112.7	26.0	116.0
			31.0	73.7	2.1	2.2	29.5	75.3	28.3	76.6	27.1	77.9	25.6	79.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.4	65.8	66.3	63.7	58.6	54.4	50.6	49.7	50.5	53.0	55.9	59.2	62.3
	DBStd	7.82	6.68	6.21	6.06	5.07	4.02	3.53	3.38	3.71	4.60	5.70	6.13	6.53
	HDD50	145	0	0	0	1	5	33	47	38	15	5	1	0
	HDD65	3088	66	49	100	204	329	431	475	450	362	292	197	133
	CDD50	2864	490	457	423	258	142	52	37	53	104	188	278	382
	CDD65	334	90	87	58	12	0	0	0	0	2	10	24	51
	CDH74	3499	891	756	546	128	3	0	0	0	19	192	332	632
	CDH80	1570	455	347	232	30	0	0	0	0	2	61	128	315
(14) Wind	WSAvg	9.2	10.1	9.6	9.0	8.0	8.3	8.2	9.3	9.6	9.6	9.6	9.5	9.9
(15) Precipitation	PrecAvg	22.6	1.5	1.4	1.5	1.8	2.0	1.8	1.9	2.1	2.1	2.4	1.9	1.8
	PrecMax	35.7	5.2	7.6	4.8	5.7	3.9	4.2	4.3	3.9	4.1	5.8	6.0	4.8
	PrecMin	11.1	0.0	0.0	0.2	0.3	0.3	0.4	0.4	0.1	0.6	0.2	0.2	0.1
	PrecStd	5.0	1.1	1.7	1.0	1.2	1.1	0.9	0.9	0.9	0.9	1.3	1.2	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	101.8	97.5	94.8	84.1	73.0	63.8	63.6	68.8	76.5	87.3	90.9	98.1
		MCWB	68.6	67.3	66.6	60.7	57.8	55.8	53.5	56.5	59.1	61.1	65.0	65.8
	2%	DB	91.9	90.5	86.4	77.4	68.1	60.8	60.1	63.8	70.2	79.2	83.3	89.1
		MCWB	67.2	66.3	65.5	60.2	57.3	54.4	52.4	53.5	57.0	59.4	64.0	64.9
	5%	DB	83.7	83.0	79.9	72.4	64.7	58.8	57.9	60.4	66.2	72.9	76.3	80.6
		MCWB	66.2	66.4	64.1	59.8	56.2	53.4	51.5	52.5	55.9	58.6	62.5	63.9
	10%	DB	76.6	76.9	73.8	68.1	61.9	57.1	56.0	57.8	62.4	67.2	70.5	73.7
		MCWB	65.5	65.6	63.5	58.9	55.6	52.5	50.8	51.3	54.2	57.1	61.0	63.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.2	72.1	69.9	64.7	61.5	58.2	55.7	57.9	61.7	64.6	69.5	70.3
		MCDB	83.2	83.3	81.9	74.9	65.2	60.8	59.8	65.3	70.0	75.8	80.5	82.8
	2%	WB	70.2	70.1	67.8	62.8	59.2	55.8	53.9	55.1	58.9	62.0	66.2	67.4
		MCDB	82.7	80.3	78.5	70.3	64.9	58.9	57.7	60.8	67.1	72.6	77.6	80.2
	5%	WB	68.0	68.3	66.2	61.4	57.6	54.3	52.6	53.4	56.9	59.8	63.7	65.3
		MCDB	79.7	77.0	75.1	68.8	62.8	57.7	56.3	58.7	64.1	69.7	73.3	76.7
	10%	WB	66.1	66.6	64.4	59.9	56.2	53.0	51.4	52.0	54.9	57.7	61.5	63.4
		MCDB	76.3	75.4	72.5	67.0	60.9	56.4	55.1	56.8	61.2	66.0	69.7	73.0
(35) Mean Daily Temperature Range	5% DB	MDBR	18.4	17.8	17.6	17.2	14.2	12.7	12.6	14.3	16.5	18.5	18.2	19.3
		MCDBR	33.3	31.0	29.4	25.6	18.9	15.1	14.7	18.8	23.2	30.8	30.7	34.1
		MCWBR	11.0	10.5	10.7	11.3	10.4	9.9	8.9	10.6	12.1	13.5	13.0	12.3
	5% WB	MCDBR	25.6	22.6	22.8	20.8	16.1	13.6	13.0	16.8	19.6	24.5	25.3	26.1
		MCWBR	10.6	9.9	9.8	10.6	10.1	9.8	8.8	10.5	11.5	12.7	12.4	11.3
(40) Clear-Sky Solar Irradiance	taub		0.352	0.355	0.335	0.327	0.314	0.301	0.303	0.312	0.333	0.339	0.347	0.343
	taud		2.509	2.510	2.565	2.564	2.574	2.602	2.587	2.558	2.489	2.488	2.493	2.525
	Ebn at Noon		311	302	296	279	262	258	264	279	292	304	310	315
	Edh at Noon		36	34	30	26	22	20	22	25	31	34	36	35
(44) All-Sky Solar Radiation	RadAvg		2247	1964	1544	1078	721	567	635	892	1278	1704	1984	2233
	RadStd		135	89	88	58	35	41	35	49	79	125	115	130

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.24	+0.81	+1.08	N/A	+0.95	+1.17	N/A	-356	+437	+95	
(47)	Regional (36 neighbors)	+0.88	N/A	N/A	+1.51	N/A	N/A	-27	-232	+311	+97	

Nomenclature: See separate page